

LAB ASSIGNMENT - 2.2

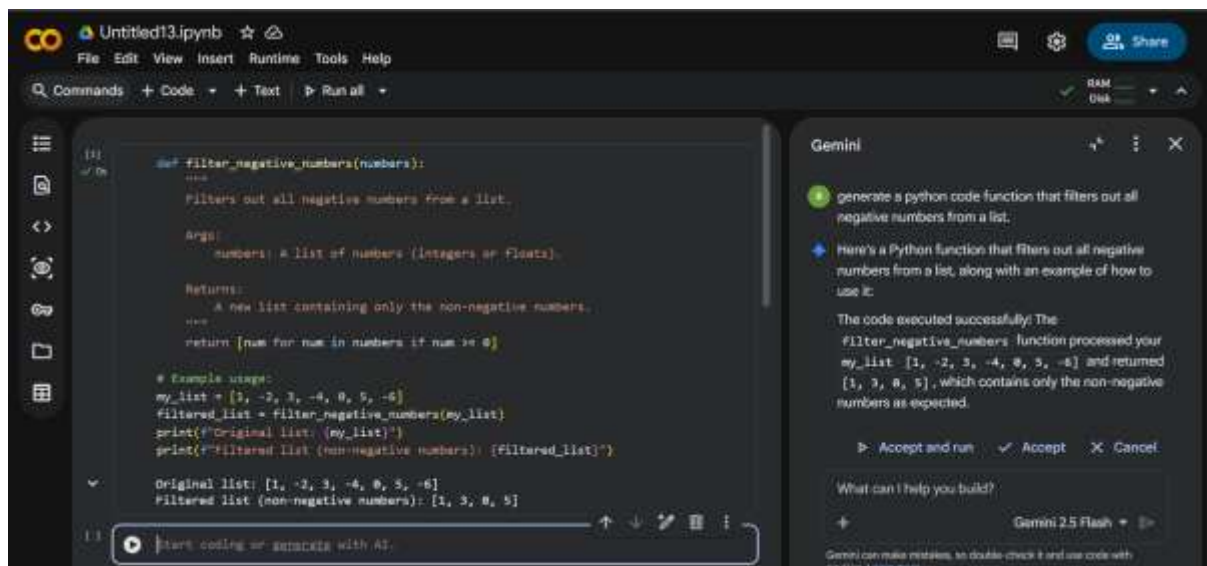
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BATCH 29

TASK-1:

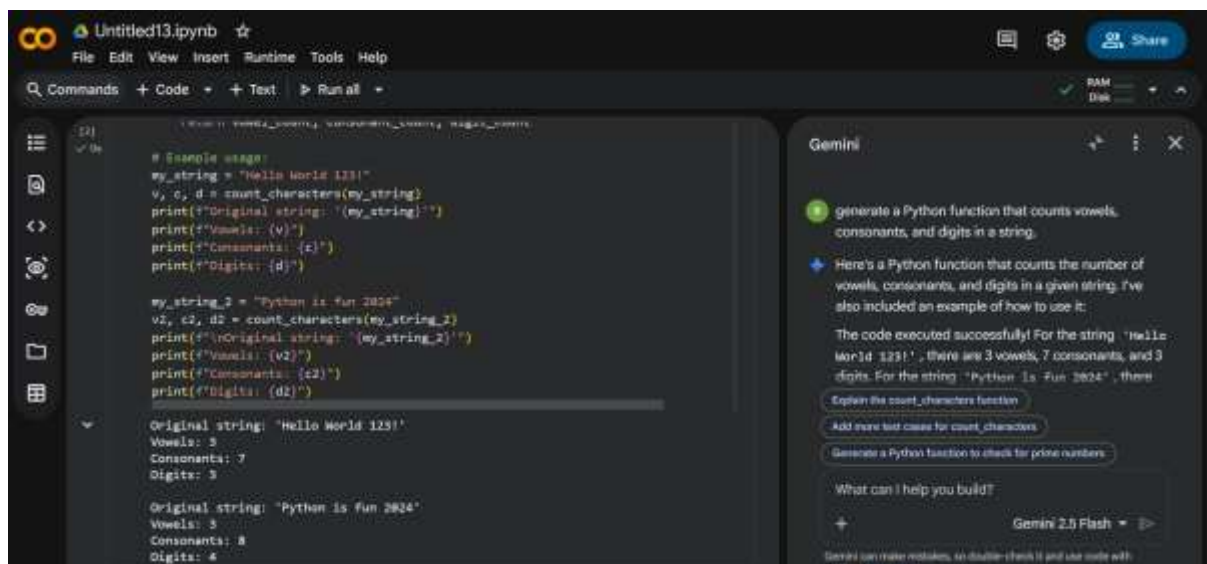
Prompt : Generate a python code for function that filters out all negative numbers from a list



```
[1]: def filter_negative_numbers(numbers):  
    """  
    Filters out all negative numbers from a list.  
  
    Args:  
        numbers: A list of numbers (integers or floats).  
  
    Returns:  
        A new list containing only the non-negative numbers.  
    """  
    return [num for num in numbers if num >= 0]  
  
# Example usage:  
my_list = [1, -2, 3, -4, 0, 5, -6]  
filtered_list = filter_negative_numbers(my_list)  
print(f"Original list: {my_list}")  
print(f"Filtered list (non-negative numbers): {filtered_list}")  
  
Original list: [1, -2, 3, -4, 0, 5, -6]  
Filtered list (non-negative numbers): [1, 3, 0, 5]
```

TASK-2:

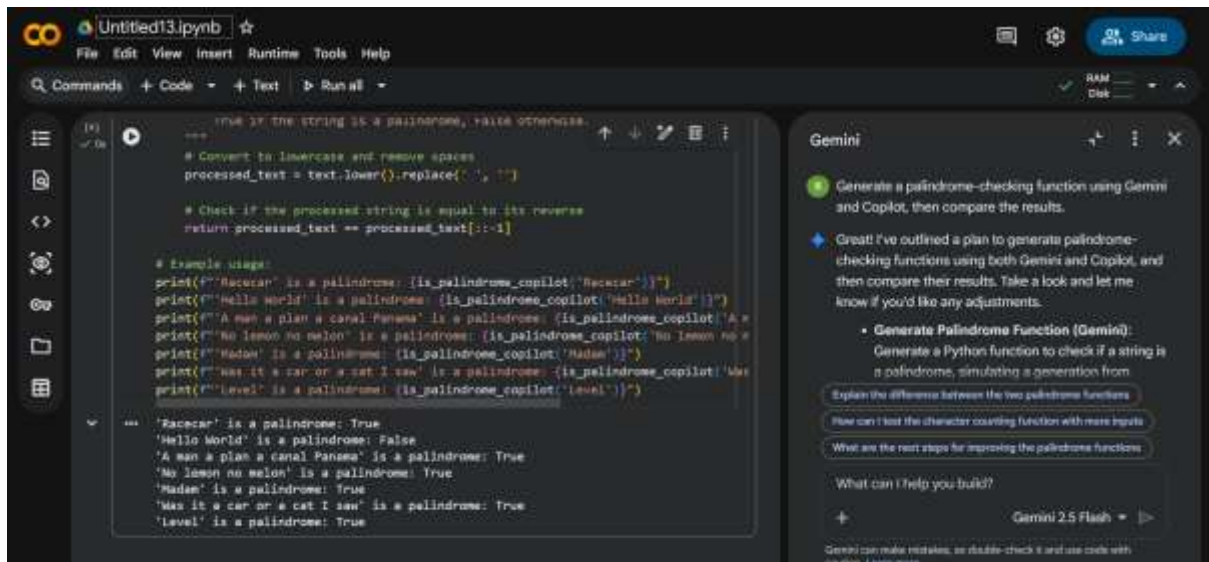
Prompt : generate a Python function that counts vowels, consonants, and digits in a string.



```
[2]: # Example usage:  
my_string = "Hello World 123!"  
v, c, d = count_characters(my_string)  
print(f"Original string: '{my_string}'")  
print(f"Vowels: {v}")  
print(f"Consonants: {c}")  
print(f"Digits: {d}")  
  
my_string_2 = "Python is fun 2024"  
v2, c2, d2 = count_characters(my_string_2)  
print(f"Original string: '{my_string_2}'")  
print(f"Vowels: {v2}")  
print(f"Consonants: {c2}")  
print(f"Digits: {d2}")  
  
Original string: "Hello World 123!"  
Vowels: 5  
Consonants: 7  
Digits: 3  
  
Original string: "Python is fun 2024"  
Vowels: 5  
Consonants: 8  
Digits: 4
```

TASK-3:

Prompt : Generate a palindrome-checking function using Gemini and Copilot, then compare the results.



TASK-4:

Prompt: explain a Python function (prime check OR palindrome check) line by line.

